

Longitudinal Associations Between Physical Activity and Sleep Duration in Infants and Toddlers

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Purpose: This study examined longitudinal associations between average physical activity (PA) levels in children and their sleep duration, and whether changes in PA levels are associated with their sleep duration. **Methods:** Data were collected on 108 children at 4 time points: when children were 6, 12, 18, and 24 months of age (44% female, 50% Non-Hispanic White). PA was assessed using accelerometry. Children's daytime, nighttime, and 24-hour sleep duration were measured with actigraphy. Linear mixed model analyses estimated the associations between average PA levels over time and changes in PA over time, treating each sleep duration variable as an outcome in separate linear mixed model analyses. **Results:** Children with higher total PA levels slept less during the day compared with children with lower total PA levels over the 2-year period. The strength of the relationship between a child's PA levels and their 24-hour sleep duration decreased as they approached 24 months of age. **Conclusions:** The results suggest that while PA may be developmentally beneficial overall, it appears that its relationship with sleep duration is not clinically relevant in very young children.

Keywords: accelerometry, actigraphy

Key Points

- While physical activity may be developmentally beneficial overall, the results of this study suggest that its relationship with sleep duration is not clinically relevant in very young children.

Sleep plays a critical role in children's health and development, although the strength of the relationship varies depending on the outcome of interest and the child's age (6). Sleep duration in early childhood can be influenced by a number of factors, including genetic, social, and behavioral factors, as well as age. As children develop physically, their sleep duration appears to decrease (5). Pediatric sleep duration guidelines recommend that infants between the ages of 4 and 12 months sleep 12 to 16 hours/day, and children ages 1–2 years sleep 11 to 14 hours/day (25). These values are associated with optimal health outcomes in these age groups, although it is important to note that the optimal sleep duration can vary between individuals. Furthermore, the wide range of values for each age group suggests a U-shaped relationship between sleep and health outcomes, with both very low and very high amounts showing a negative association with health (5). As of 2019, 80% to 90% of children in the United States were meeting sleep guidelines. However, these values vary significantly depending on household income (18). This may explain, in part, why research among older children shows that children from low-income populations are at a higher risk of negative health outcomes associated with poor sleep (12,21).

Although surveillance data suggest that most infants and very young children are sleeping an average of over 12 hours/day, it is

important to note that most of these data are collected using parent-reported sleep (10). Parent-reported sleep data are subject to social desirability and recall bias, potentially leading to overestimation of sleep (27). To overcome this issue, actigraphy is a commonly used, device-based measure of sleep (31). Actigraphy allows for the objective measurement of whether an individual is asleep or awake in a natural setting over an extended period and has been validated against polysomnography, the gold standard for sleep assessment, in very young children (31,35). Ambulatory Monitoring Inc (AMI) actigraphs are the most commonly used devices in pediatric sleep research, and the Sadeh et al (32) algorithm is the most commonly applied algorithm when scoring sleep measured with AMI devices (33).

Physical activity (PA) includes all bodily movements that lead to energy expenditure beyond that of resting levels (4). PA is considered a noninvasive and inexpensive method for improving sleep, specifically by increasing sleep duration (7). There are a number of mechanisms that have been hypothesized to explain the positive association between PA and sleep duration, although this line of research has been conducted primarily on adolescent and adult populations (1). For example, PA is thought to influence an individual's circadian rhythms, change hormonal levels, affect body temperature and metabolism, and alter heart rate and mood, all of which can result in changes to sleep (1). To better understand whether these mechanisms exist in younger populations, the research must first understand whether similar associations between sleep and PA exist across various developmental stages.

The research assessing the relationship between sleep and PA in children <3 years old is limited (16). Even fewer studies have

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assessed this relationship using device-based measures of both sleep and PA. Overall, a majority of the research is cross-sectional (3,8,13,24,39) and shows mixed results, suggesting positive but weak (3,8,13,24,39) or null (1) associations. Very few studies have assessed this relationship longitudinally among children under the age of 6 years, and there have been no longitudinal studies conducted on the sleep and PA relationship in infants (16,37). Longitudinal data analysis is a beneficial method for assessing the sleep and PA relationship because it allows for the disaggregation of within- and between-person effects of PA over time. These methods determine the extent to which changes in an individual's PA levels (the within-person effect) are associated with sleep, as well as how differences between subjects' PA levels (the between-subject effect) are associated with sleep, while accounting for other variables involved in children's development that are associated with sleep (15). Only one study, which resulted in null findings, has explored the within- and between-person associations of PA and sleep among 3- to 6-year-old children (36).

Although longer sleep duration and better sleep quality are associated with higher levels of PA among older populations (17), the current research that exists in infants and young children shows mixed results. In fact, the 2018 Physical Activity Guidelines Advisory Committee Scientific Report concluded that "Insufficient evidence is available to examine relationships between PA and sleep in children and adolescents . . ." (29). Therefore, this study will determine whether there is a longitudinal relationship between device-based measures of sleep duration and PA in a diverse sample of children 6 to 24 months old. Specifically, this study aims to: (1) examine whether PA levels in 6- to 24-month-old children, measured as an average over the 2-year measurement period, are associated with sleep duration and (2) examine whether changes in PA levels in children from 6 months old to 24 months old are associated with sleep duration.

Participants and Methods

Participants

Data for these analyses were from the Linking Activity, Nutrition and Child Health (LAUNCH) study. The LAUNCH study protocol can be found elsewhere (26). This study was approved by the Institutional Review Board of the University of *blinded*. The LAUNCH study collected longitudinal, observational data in children at 6 time points, at ages 6, 12, 18, 24, 30, and 36 months. The present study focused on the associations between sleep and PA using the first 24 months of data. Participants were 108 children, and their biological mothers recruited from communities in South Carolina. The LAUNCH study excluded children with any physical limitation that prevented the use of accelerometry to assess PA, as well as children born before 37 weeks gestation.

Measures

Sleep

Sleep was measured using actigraphy (MicroMini Motionlogger, AMI). Parents of children enrolled in the study received actigraphs during each data collection visit with parents at his/her home, childcare center, or another predetermined location. Children wore actigraphs on the left ankle for 24 hours/day for 7 days, except during water-based activities. Data were assessed using software available from the manufacturer (Action4, version 1.16, AMI). Sleep diaries were not used alongside actigraphy in this study to

avoid a loss of data resulting from noncompliance with sleep diary completion and to reduce participant burden (38).

Sleep duration variables were computed using the Sadeh et al (32) algorithm and assessed in 60-second epochs. Wear time was determined using the LIFE channel included in the manufacturer-provided software (MicroMini Motionlogger, AMI) (34). Participants were required to have ≥ 3 days/nights with ≥ 10 hours of data/night to be included in analyses assessing daytime sleep duration and nighttime sleep duration. Each participant needed ≥ 3 24-hour days with ≥ 20 hours of data/night to be included in analyses assessing 24-hour sleep duration.

Nighttime sleep duration was operationalized as the number of minutes spent sleeping between 7 PM and 7 AM, averaged over the 7-day assessment period at each of the 4 time points. Daytime sleep duration was operationalized as the average number of minutes spent sleeping between 7 AM and 7 PM. These definitions of nighttime and daytime sleep were based on research suggesting that very young children have an average bedtime between 6:30 PM and 9:00 PM and an average wake time from 6:00 AM to 8:00 AM (22). A wakeup time of 7 AM and a bedtime of 7 PM were within this window of time. Twenty-four-hour sleep duration was operationalized as the average number of minutes spent sleeping between 7 AM and 7 AM the following day. This approach for determining daytime and nighttime sleep was also taken because the sleep and PA monitors were not worn simultaneously for all participants at all time points. Therefore, determining sleep duration individually based on each participant's bed and wake time could potentially result in "days" that did not total 24 hours. The daytime and nighttime sleep duration variables were normalized by being divided by wear time and extrapolated into 12 hours. The 24-hour sleep duration variable was normalized by being divided by wear time and extrapolated into 24 hours.

Physical Activity

ActiGraph accelerometers (GT3X model) were used to collect PA activity data. Parents of children enrolled in the study received accelerometers during each data collection visit with parents at his/her home, childcare center, or another predetermined location. An accelerometer was worn for 7 days and was placed inside a child-sized sweatband and worn on the right ankle. Trained data collectors provided caregivers with instructions on how and when the monitor should be worn. The monitor was removed during water-based activities and at night and attached shortly after waking each morning. Accelerometers were initialized before collecting data at midnight following accelerometer distribution.

Data were collected in raw data mode at 80 Hz and later downloaded into 15-second intervals (2). Nonwear time was defined as any ≥ 60 -minute period of consecutive zeros and was excluded from analyses (28). Missing PA data were imputed for children who had >1 day of >8 hours of accelerometry data between the hours of 7 AM and 7 PM. This was done using the PROC MI command with 20 imputations to predict missing vector magnitude (VM) count values for each day. There were 150 children originally enrolled in the LAUNCH study. There were 130 children at time point 1, 102 children at time point 2, 84 children at time point 3, and 44 children at time point 4 with >1 day of >8 hours of accelerometry data. Predictor variables used for imputation included the 7 days of VM counts and the number of hours of wear time for each of the 7 days. The mean VM counts over 7 days were averaged for each child after imputation. Total PA counts were operationalized as the average number of VM counts/hour between 7 AM and 7 PM each day. This PA variable was

chosen because intensity cut points have not yet been established for nonambulatory children (2).

Demographic Characteristics

Demographic characteristics were reported by the child's mother at baseline. These include sex and race/ethnicity. Socioeconomic status (SES) was assessed by asking whether the mother was enrolled in the Special Supplemental Nutrition Program for Women, Infants, and Children (WIC) and was collected at each time. Individuals are eligible for WIC in South Carolina if they fall below 185% of the federal poverty guidelines.

Statistical Analyses

All analyses were conducted using SAS (version 9.4). Skewness assessed the normality of the PA data and showed that PA data were normally distributed. Continuous variables (age, total PA, nighttime, daytime, and 24-h sleep duration) were summarized using means and SDs, and percentages were used to summarize categorical variables (sex, race, and WIC status) at baseline. All models were likelihood based and used all observed data, resulting in validity under the missing at random assumption (19).

The average daytime PA VM counts/hour variable was treated as the independent variable of interest. We estimated the separate associations of between- and within-person changes in PA with the sleep outcomes. The between-person variable modeled the overall association for PA and sleep between different children, while the within-person variable modeled how changes in PA over time within a child (eg, from 6 to 12 mo) were associated with sleep. The between-person variable was calculated by taking the average of an individual child's PA data from all 4 time points. The within-person variable was calculated as the difference between a child's average PA at a specific time point subtracted from that child's overall average PA. To ensure the regression coefficients for the PA and sleep variables had similar scaling, all PA variables were expressed as counts/minute/1000. Sensitivity analyses were conducted to assess for collinearity between age and change in PA. Results indicated that collinearity had minimal impact on these variables.

Unadjusted and adjusted linear mixed models were analyzed using the PROC MIXED function, which can be applied to longitudinal studies with repeated measures. Separate models were created where daytime sleep duration, nighttime sleep duration, and 24-hour sleep duration were each used as the dependent variable. Variables in the models with a P value $\leq .05$ were considered statistically significantly associated with sleep.

Time was operationalized as the child's time-varying mean-centered age (cluster-centered, in months). Models included both the average PA variable (the between-person PA variable) and the individually mean-centered PA variable (the within-person PA variable) as independent variables. A mean-centered age by mean-centered PA interaction was included in each of the models to determine whether the association between sleep and PA changed with age. An unadjusted model with baseline random intercepts was compared with a model with random intercepts and slopes for each analysis. The best-fitting model was chosen based on the smallest Akaike information criterion. Residual plots were created to check the assumptions of all models, and no violations were found. Adjusted models followed the same format as unadjusted models, with the addition of adjustments for the child's race/ethnicity (Non-Hispanic Black, Non-Hispanic White, Hispanic, Other), sex (male, female), and WIC enrollment (yes, no).

Results

Preliminary Analyses

A total of 150 children were enrolled in the study. Observations that did not meet the inclusion criteria for sleep were removed, resulting in a sample size of 109 children with sleep data at ≥ 2 time point(s): 87 children had sleep data at time point 1, 71 at time point 2, 56 at time point 3, and 55 at time point 4. One child was excluded due to missing data on WIC enrollment. This resulted in a sample size of $N = 108$.

Descriptive Statistics

Baseline measures showed that a majority of the children (50%) were Non-Hispanic White, followed by Non-Hispanic Black (27%), Hispanic (17%), and Other (6%). Just over half of the children were male (55%) and 45% were female, and 62.8% of children were from families not receiving WIC assistance. Table 1 provides means, SDs, and ranges for children's ages, sleep, and PA data at each time point.

Unadjusted Models

In unadjusted analyses, the between-subject effect of PA measured at the ankle was significantly, positively associated with nighttime sleep duration ($P < .01$). Based on unadjusted analyses, children who were more physically active slept longer at night compared with children who were less physically active, but this association was no longer significant in the adjusted model.

Adjusted Models

Table 2 shows adjusted models assessing the relationship between sleep and PA measured using accelerometry at the ankle. Unlike in the unadjusted models, the between-subject effect of PA was negatively associated with daytime sleep duration ($P = .04$). Results indicated that 1000 more PA counts per hour corresponded with 53 fewer seconds of sleep during the day. The interaction between age and the within-subject effect of PA was negatively associated with 24-hour sleep duration, and this relationship did not achieve statistical significance ($P = .07$). Non-Hispanic Black children slept nearly 37 minutes less at night on average, compared with Non-Hispanic White children ($P = .01$). Children whose mothers were enrolled in WIC slept less at night compared with children whose mothers were not enrolled in WIC ($P < .01$).

Discussion

This study examined the longitudinal relationship between PA and sleep in children under 2 years of age, and results indicated that 1000 more PA counts/hour corresponded with 53 fewer seconds of sleep during the daytime. It is important to note that although the relationship between sleep and PA was significant, the magnitude of this relationship may not be clinically relevant. Although other studies have found negative associations between sleep and PA in very young children (39,42), these associations were seen in the nighttime (42) and 24-hour sleep period (39) and not in the daytime sleep period. These differences can be due to a number of methodological differences, including the location of accelerometers (42), the use of a singular device to measure both sleep and PA (39), the epochs used for sleep and PA analysis (39,42), and the

Table 1 Descriptive Statistics of Sleep and Physical Activity Data at Each Time Point

	Time point 1			Time point 2		
	N (%)	Mean (SD)	Range	N (%)	Mean (SD)	Range
Age, mo	87	6.93 (1.10)	5.00–10.24	71	12.78 (1.01)	10.53–16.16
Sleep						
24-h sleep duration, min	87	732.13 (111.67)	443.33–1289.20	71	739.38 (67.97)	597.00–926.25
Daytime sleep duration, min	87	189.13 (94.07)	74.50–685.80	71	185.53 (53.82)	59.75–328.43
Nighttime sleep duration, min	87	545.48 (75.88)	319.00–687.83	71	557.96 (66.66)	393.80–654.50
Physical activity						
Ankle VM CPH/1000 (7 AM–7 PM)	84	80.44 (25.78)	31.40–166.67	60	91.77 (21.16)	41.49–148.99
	Time point 3			Time point 4		
	N (%)	Mean (SD)	Range	N (%)	Mean (SD)	Range
Age, mo	56	19.60 (1.52)	17.45–22.65	55	25.53 (1.53)	23.54–29.00
Sleep						
24-h sleep duration, min	56	738.26 (99.58)	594.67–1292.33	55	725.85 (69.35)	525.00–865.67
Daytime sleep duration, min	56	194.09 (124.08)	82.17–879.83	55	166.91 (52.28)	43.83–327.40
Nighttime sleep duration, min	56	551.28 (87.71)	317.00–679.40	55	562.47 (60.57)	424.17–684.17
Physical activity						
Ankle VM CPH/1000 (7 AM–7 PM)	56	109.19 (24.82)	59.53–177.40	30	125.74 (39.01)	52.41–227.09

Abbreviations: CPH, counts per hour; VM, vector magnitude.

Table 2 Associations Between PA Measured at the Ankle and Sleep Duration,^a Adjusted for Independent Factors

	24-h sleep duration		Daytime sleep duration ^b		Nighttime sleep duration	
	Estimate (SE)	P	Estimate (SE)	P	Estimate (SE)	P
Intercept	772.94 (35.74)	<.001	224.64 (30.79)	<.001	564.50 (27.89)	<.001
PA—b/w-subject effect	−0.20 (0.30)	.52	−0.53 (0.27)	.04	0.26 (0.24)	.29
PA—w/in-subject effect	−0.58 (0.34)	.10	−0.53 (0.40)	.18	0.10 (0.19)	.61
Age ^c	0.77 (1.08)	.48	1.06 (1.21)	.38	0.09 (0.77)	.90
PA ^d × age ^c	−0.02 (0.05)	.75	0.03 (0.05)	.62	−0.05 (0.03)	.07
Race/ethnicity						
Non-Hispanic White	Ref	—	Ref	—	Ref	—
Non-Hispanic Black	7.53 (21.62)	.73	28.00 (17.33)	.11	−35.93 (15.64)	.03
Hispanic	21.65 (24.64)	.39	32.14 (9.53)	.10	−21.70 (17.49)	.22
Other	−41.47 (36.71)	.27	43.51 (27.44)	.12	40.89 (27.23)	.14
Sex						
Male	Ref	—	Ref	—	Ref	—
Female	−15.15 (16.23)	.36	−13.19 (12.57)	.30	−9.36 (11.56)	.42
WIC						
Not enrolled	Ref	—	Ref	—	Ref	—
Enrolled	−35.72 (18.71)	.06	15.42 (15.81)	.33	−45.24 (12.50)	<.01

Abbreviations: b/w, between; PA, physical activity; Ref, reference category; w/in, within; WIC, Women, Infants and Children Nutrition Program. Note: (N=108).

^aNighttime and daytime sleep duration and number of nighttime awakenings were normalized by dividing values by wear time and multiplying by 12. 24-hour sleep duration was normalized by dividing values by wear time and multiplying by 24. ^bModel includes random intercept only. ^cMean centered. ^dWithin-subject effect of physical activity.

definition on nocturnal sleep and daytime naps (42). A limited number of studies have found that higher PA levels were associated with longer sleep duration in very young children (8,14), although they were limited by the use of parent-reported sleep data (8) and by not adjusting for the demographic characteristics (14). The present study expands on previous research by examining these

relationships over an 18-month period and by using device-based measures.

The inverse association seen in the present study may be due to time constraints. Because there is a limited amount of time in the day, performing 1 activity displaces the time a child has to perform another. In other words, children who sleep more during the day

may have less time to be physically active, and vice versa. New data analysis techniques are being used to account for this constraint (20), but they require data that differentiate between sedentary behavior and PA. The use of accelerometry to measure PA has not been done extensively in very young children, and PA intensity cut points have not yet been established for preambulatory children. This means that a clear and consistent distinction between sedentary behavior and light-intensity PA cannot be made. Therefore, these new data analysis techniques cannot be conducted in infants and very young children until PA intensity cut points have been established.

The statistical approach used in the present study allowed for the assessment of how intraindividual changes in PA across 24 months were associated with sleep. There were no significant associations between intraindividual changes in PA across 24 months and children's sleep durations. The methodology applied to the present study has only been used in one other study assessing sleep and PA in very young children, but resulted in null findings (36). Furthermore, it differed in its methods from the present study in that it was microlongitudinal and only assessed the repeated measures over a 1-year period (36). The present study expanded on the limited previous research by assessing the sleep and PA relationship over 18 months rather than over a week. The negative sleep and PA association in this study is inconsistent with the findings seen in older children and adults, which suggest a positive relationship between the 2 behaviors (17).

Multiple mechanisms have been used to explain the relationship between sleep and PA in older children. For example, PA may improve an individual's circadian rhythm; lower resting heart rate; alter the rate of growth hormone secretion while an individual is sleeping; and improve an individual's metabolism, immune system, and mood, all of which could lead to better sleep. These mechanisms might not be relevant when exploring the sleep and PA relationship in younger children due to major developmental differences between very young children and older children or adolescents (1,36). As children transition from infancy, into toddlerhood, and then into childhood, they also transition from a polyphasic, to a biphasic, and ultimately to a monophasic sleep pattern (10). Considering sleep patterns undergo multiple changes during this transition, it is possible that the relationship sleep has with other health behaviors, such as PA, also changes (1,37). Additionally, very young children have very little control over the timing of activities, including bedtime and time for PA. The previously proposed mechanisms that exist in older children and adults may not appear until children begin to gain some autonomy over their schedule (1). This argument is supported by the fact that the interaction between age and the within-subject effect of PA was marginally, negatively associated with nighttime sleep duration. In other words, as children approached 24 months of age, the association between their 24-hour sleep duration and their PA levels decreased in strength. As such, researchers should avoid applying mechanisms developed in older children and adults to infants and very young children, and should instead consider whether there are other biopsychosocial mechanisms underlying the sleep and PA relationship in infancy and young childhood.

Differences in sleep duration were seen across race/ethnicity and SES groups. Non-Hispanic Black children slept 40 minutes less than Non-Hispanic White children, and children from lower SES households slept 47 minutes less than children from higher SES households. These results are consistent with data from the 2016–2018 National Survey of Children's Health, which showed

that the prevalence of short sleep was highest among Black children and families with lower income (40). There is a large amount of evidence pointing to various social and environmental factors related to sleep outcomes. This included cultural practices surrounding sleep, such as bed sharing (11), parental exposure to discrimination, which can negatively affect mother-infant interactions and cause a disruption in bedtime routines (30), and environmental factors in the home and in the neighborhood, such as light and noise pollution (11). Developing tailored interventions that address the individual-level behaviors of parents to improve their child's sleep is not sufficient, and more work is needed to address the large-scale social determinants of health that are directly related to sleep to improve infant and toddler sleep in at-risk populations.

The major strength of this study was the use of device-based measures for sleep and PA in a longitudinal study with a diverse population of infants and toddlers (1). Furthermore, this study adds to the literature by using a unique longitudinal analytic method. This method assessed how differences between children's PA levels along with changes in individual children's PA levels were associated with sleep over time. Previous research has primarily focused on assessing differences between children's PA levels and their association with sleep over time (15).

Limitations to the study must also be addressed. The use of accelerometry in this age group can be affected by caregiver handling (41). The use of actigraphy is also limited in this age group, as it is not very accurate at discriminating between sleep and wake classifications when the child is inactive during a wake period or moving while asleep (9). This study did not ask parents to keep sleep diaries, which is recommended (38), and could have assisted in the notation of days or nights that are not necessarily representative of a child's actual sleep behaviors, as well as better differentiating between sleep and sedentary time. The present study defined daytime sleep as any minute scored as sleep by the algorithm that occurred between 7 AM and 7 PM. Other methods have been developed to score daytime napping, although there is no consensus at this time as to a standard approach for determining daytime napping. Therefore, the researchers for the present study used the same methods for scoring daytime and nighttime sleep duration. Furthermore, this study did not address the relationship between sleep quality and PA in this age group, although more research is needed to clearly identify measures of sleep quality in this age group before assessing its relationship with PA (23).

As more work is conducted on ways to improve sleep, it is important to understand whether PA can be used in multicomponent interventions targeting improvements in sleep outcomes among infants and young children in the same way it has been used in adults. The few interventions that have been conducted in younger children have resulted in null findings (37), suggesting that although PA may be beneficial for other health outcomes such as the development of healthy weight status and motor development in very young children (6), it may not be associated with sleep at this developmental stage. Future research should assess whether the sleep and PA relationship varies depending on PA intensity once established cut points are developed for PA in infants and very young children. Additionally, this study assessed the relationship between sleep and the within- and between-person effects of PA, but the relationship between PA and the within- and between-person effects of sleep was not assessed. Previous research suggests this association may exist in very young children (36), and it should be explored further in future studies.

Conclusions

The present study found that children with higher total PA levels slept less during the day compared with children with lower total PA levels. These results suggest that the mechanisms underlying the sleep and PA relationship in young children vary from those that have been suggested in older children and adults. Future research assessing the sleep and PA relationship in infants would benefit from the development of PA intensity cut points to determine whether the relationship varies depending on PA intensity.

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